Joseph Resch

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Education

University of California, Los Angeles (UCLA), Los Angeles, CA 2020 – 2026
Statistics Ph.D. | GPA: 3.94 | University Teaching Fellow

University of Georgia, Athens, GA 2016 – 2020
Statistics B.S. | Economics B.A. | Undergraduate Research Fellow

Skills

Programming: Python, R, SQL, Bash, GitHub, AWS (SageMaker, S3), Tableau, C++, SAS, JMP
ML Libraries: Pandas, NumPy, scikit-learn, PyTorch, Optuna, SHAP, Matplotlib, Shiny, Rcpp
Statistical Methods: Machine Learning, Bayesian Modeling, Hierarchical Modeling, Latent Variable Models, Variational Inference, MCMC, Causal Inference, A/B Testing, Experimental Design, Segmentation, Simulations

Experience

Doctoral Researcher, Statistics 2020 – 2026
UCLA Statistics & Data Science

- Advanced statistical models for social and economic network data using exponential-family random graph models (ERGMs), enabling inference on large graphs with complex structural dependencies
- Derived Bayesian reference prior for ERGMs and implemented Metropolis-Hastings MCMC and variational Bayes approaches; improved model goodness-of-fit by ~15%
- Built simulation benchmarking framework to evaluate likelihood approximation procedures
- Optimized pipeline on high-performance cluster computing for large-scale stimulation studies
- *VariationalEstimationERGM* (R Code) – github.com/josephresch/VariationalEstimationERGM

Data Science Intern 2025
Farmers Insurance

- Developed scalable gradient boosting models (XGBoost, LightGBM) to predict auto accident risk
- Engineered features from 80+ million row Snowflake SQL pipelines and trained models in AWS SageMaker with GPU acceleration
- Tuned hyperparameters using Optuna cross-validation and applied SHAP for interpretability
- Deployed modular, reusable Python workflows portable across domains (forecasting, segmentation, and behavioral analytics), improving high risk detection by 2x over existing infrastructure

Research Data Scientist 2022 – 2025
UCLA School of Education & Information Studies

- Led U.S. Department of Education (IES) research project advancing hierarchical latent-space models using multivariate logistic regression for time-series psychometric applications
- Developed Bayesian auxiliary sampling framework enabling likelihood inference using Gibbs sampling and variational inference methods via Pólya-Gamma data augmentation
- Reduced runtime by ~30% through model redesign and Bayesian sampler optimization
- Published findings and code; repo: *BayesianTDCM* (R Code) – github.com/josephresch/BayesianTDCM

Research Assistant 2018 – 2020
University of Georgia, Department of Statistics

- Optimized a time-series rainfall-runoff hydrological simulator using sequential Gaussian process surrogate modeling and Latin hypercube space-filling designs to enable downstream causal analysis
- Performed parallelized large-scale simulations on a computing cluster using R
- Improved goodness-of-fit over competing algorithm by 20%; published and presented at conference

Selected Publications

First-author publications: Psychometrika (2025) – *Psychometric model*, PLOS One (2025) – *ERGM inference*
Co-author publications: Journal of Statistical Theory and Practice (2023) – *Hydrology model*